H-Bridge

2024/2025

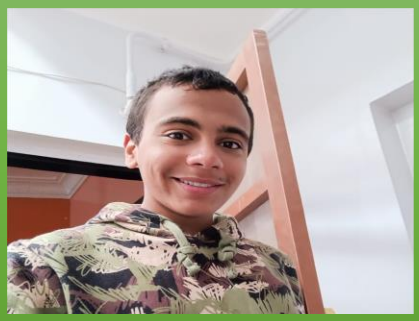
Grade 10

Semester 1

Group No.23110



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Introduction

Egypt faces several significant challenges in infrastructure and environmental sustainability, which are known as Egypt grand challenges. Addressing these issues requires a strategic approach to create a luminous future, improve citizens' quality of life, and ensure Egypt's sustainable growth.

Urban congestion

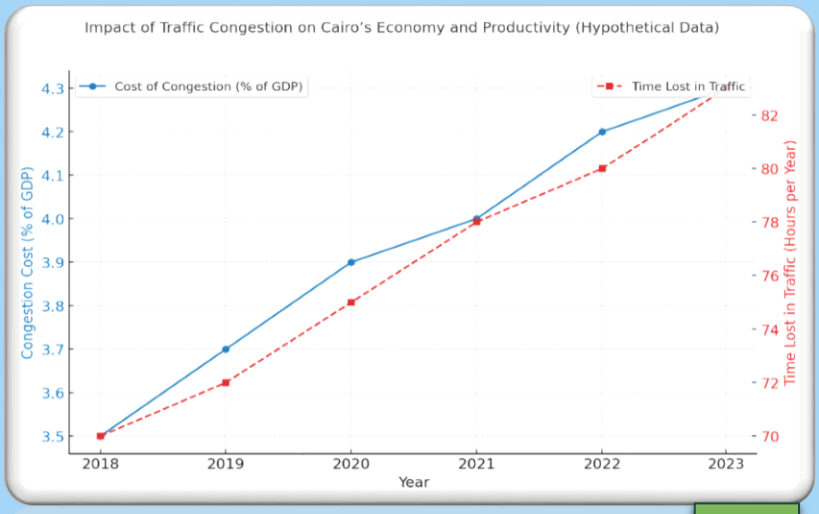


Fig .1

One of the most pressing issues Egypt confronts is its infrastructure, particularly in densely populated areas like Cairo. High traffic congestion has become a daily struggle for millions of people, with long hours lost in traffic contributing to air pollution and economic strain. The World Bank (2023) estimates that traffic congestion in Greater Cairo costs Egypt up to 4% of its GDP annually, approximately EGP 50 billion (USD 8 billion), as shown in **Fig .1** (World Bank, 2023). This impact is seen through reduced productivity, increased vehicle emissions, and overall diminished urban quality of life.

Climate change

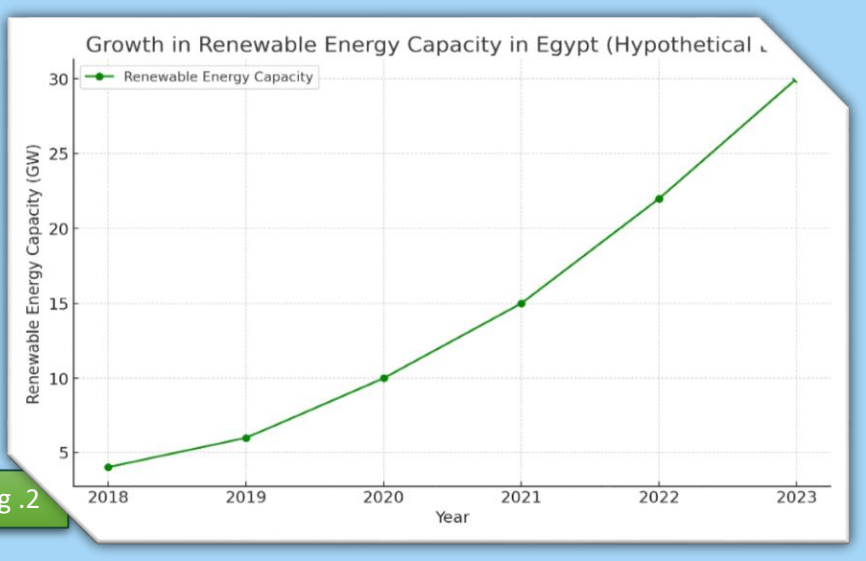


Fig .2

Alongside its urban congestion challenges, Egypt is also highly influenced to the effects of climate change, which threatens essential sectors like water resources, agriculture, and public health. Due to the lack of pure water as the climate affections, food security has become a urgent issue, especially with Egypt's rapidly growing population (Brookings Institution, 2023). As shown in **Fig .2**, the country's dependency on the Nile River for fresh water poses further risks, amplifying the need for sustainable water management solutions.

Recycling Waste: Economic and Environmental Benefits



Fig .3 **Composition of waste in Egypt**

Egypt's rapid urbanization has led to increased waste generation, creating significant economic and environmental challenges. The poor control in management of wastes not only cause the consume the public resources it's also one of the major causes of public health issues. Recycling offers an applicable solution to these problems, as it can generate economic opportunities while mitigating environmental impacts (*OpenEdition Books, 2023*). As illustrated in **Fig. 3,** implementing effective recycling programs could create jobs and reduce waste disposal costs.

Moreover, recycling plays a crucial role in conserving natural resources and reducing greenhouse gas emissions. By diverting waste from landfills, Egypt can promote a circular economy, where materials are reused and repurposed, leading to a sustainable future. Given the pressing need for efficient waste management, the integration of recycling practices is essential for both economic development and environmental preservation of Egypt.

Building Bridges to Solve Egypt's Challenges

There are a lot of ways we can tackle Egypt's big issues, and building a bridge is one solution that stands out. By creating new bridges, we can make transportation easier, connect areas that

are cut off, and even open up dry, unused land for people to live and work. Bridges help reduce traffic in crowded cities by giving people faster routes, cutting down on the time spent in traffic, and lowering pollution. With better connections to new areas, more people can move away from the packed city centers, which helps spread out the population and reduces pressure on urban areas.

Building a bridge also helps the economy by making it easier and cheaper to move goods around, which boosts trade and business. Plus, we can make bridges more environmentally friendly by adding things like solar panels to power lights or using sustainable materials. By focusing on building bridges like this, we don't just improve traffic—we also take steps toward a cleaner, greener future for Egypt.

Our Project: Creating a Bridge Prototype

In this project, our goal is to create a prototype for a bridge that could help solve these problems. We'll design a model bridge with:

- Strong support
- Efficient traffic flow
- A sustainable layout

We'll choose the best location for the bridge so that it helps reduce congestion the most, and we'll pick the right materials to make sure it's sturdy and eco-friendly.

By carefully planning every part of this bridge, we hope to create a design that could help solve Egypt's challenges in real life. This portfolio will walk through each step of building this model and explain how a well-built bridge can make life better for everyone.

Present and Justify a Problem and Solution Requirements

1. Egypt Grand Challenge(s)

I. Urban congestion

Fig .4



Despite the vast lands that Egypt holds, it still struggles with urban congestion. Urban congestion is an issue that creates problems in various aspects of life in Egypt, as illustrated in **Fig .4.**

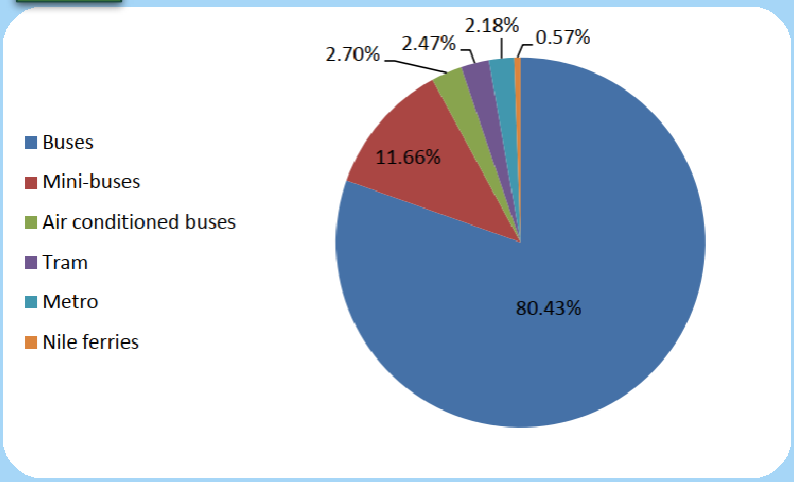
The Basic Cause of Congestion is friction among vehicles in the traffic flow or the situation that arises when the entry of one vehicle into a traffic flow increases the journey time of the rest.

How can it be addressed?

Urban congestion can be solved by various methods. One of those strategies is to utilize a bridge. Bridges help facilitate traffic by providing an alternative track over obstacles like rivers, waterways, railways, or other busy roads.

Additionally, they improve traffic flow, shorten travel time, and reduce congestion on main roads; they also supply rigid road networks, as shown in **Fig .5.**

Fig .5



(Brookings Institution, 2022). Urban congestion poses substantial challenges in rapidly urbanizing regions, such as Egypt, where increased population density exacerbates traffic delays, pollution, and stress on infrastructure systems. Sustainable urban planning and investment in public transit are essential to mitigate these issues and improve overall quality of life.

II. Improving the use of arid areas

Interest in Egypt's dry and **arid regions** has recently increased. The for more food and alternative energy resources and the population growth have encouraged the people to look for new areas where they can establish cities, buildings, and new life.

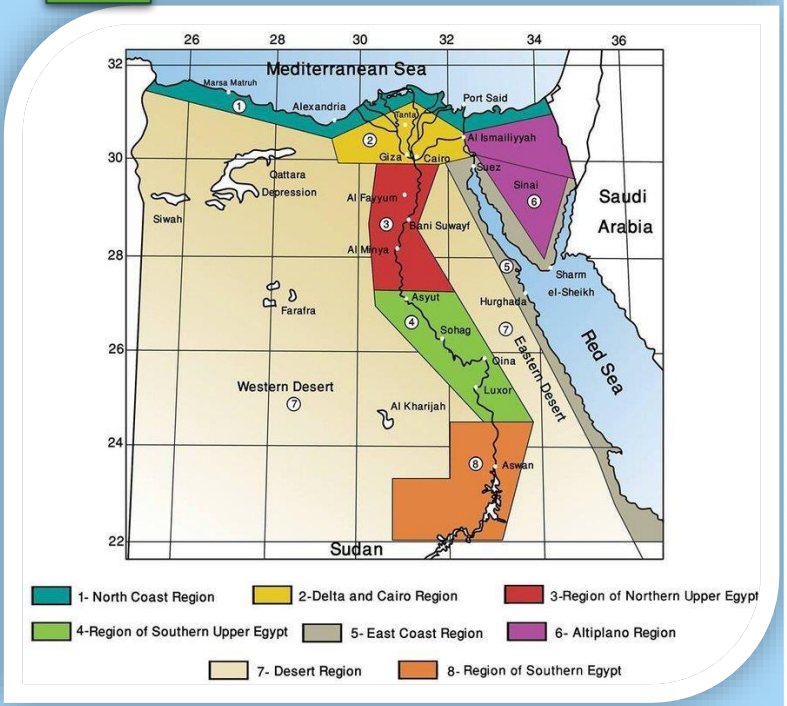
Actually, 96.0% of Egyptian land is made up of dry zones, which is a great segment of Egypt.

Why is improving the use of arid areas is important?

Improving the use of arid areas plays a preeminent role in

- Food security: As arid areas contribute in food production due to the unconsumed resources that those lands provide.
- Population distribution: As improving the use of arid a would promote the people to dwell in these areas it also provides a distribution of population that led to prevent population concentration in a specific area like capitals and coastal cities
- Economic development: As utilizing arid areas can open new opportunistic opportunities for economic growth.

Fig .6



Improving Land Use in Arid Regions: Strategies and Impact

The sustainable development of arid regions is a pressing issue, particularly in countries with expansive desert areas. Strategic land-use improvements are essential not only for supporting regional growth but also for preserving delicate ecosystems. Research from the *Journal of Geographical Sciences* underscores the importance of balancing ecological health with economic viability, recommending sustainable land-use patterns tailored to arid environments (Journal of Geographical Sciences, 2010). Furthermore, the U.S. Forest Service provides detailed insights on

landscape restoration methods that have proven effective in rehabilitating arid ecosystems (USDA Forest Service).

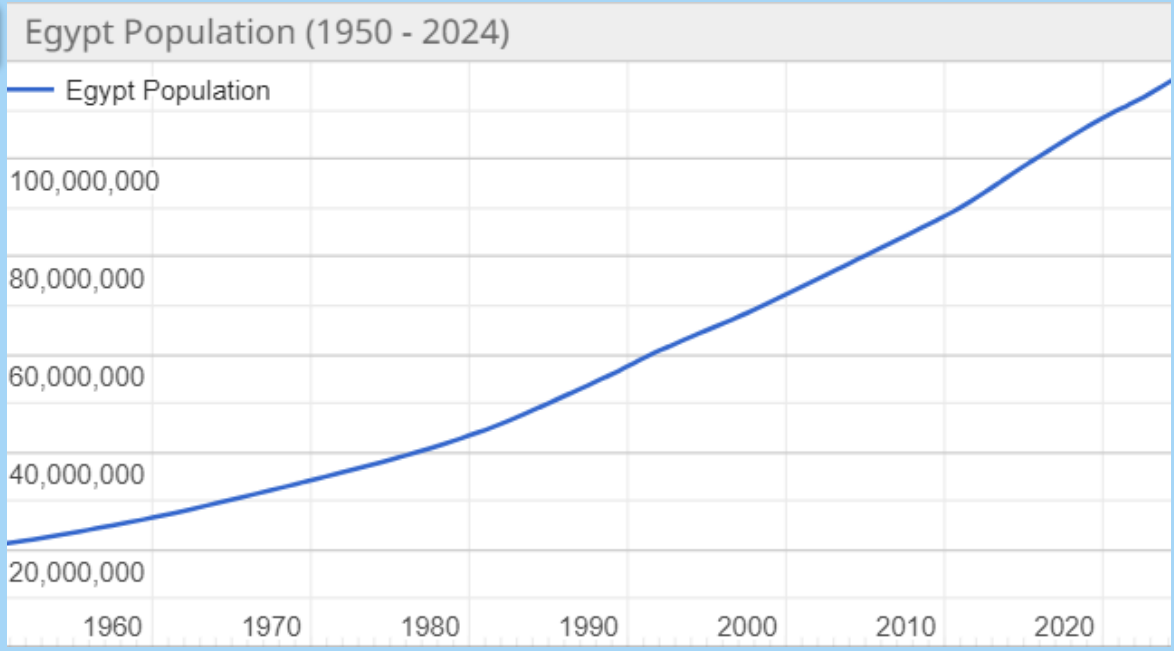
How can we use arid areas?

To employ **arid areas,** we need to establish a connection between and **urban areas** and **arid areas,** to establish this connection one of the brilliant strategies is to use bridge which can be the connection corridor between those two areas. Also make it easier to be reached.

III. Population growth and its consequences

Population growth is a critical global issue, as the increasing number of people in civil areas has driven us to higher demands on diverse resources. By observing **Fig (7)** we can notice how the Egyptian population climbed up all these million from 20 million in 1960 to 100 million in 2020 which is massive statistic.

Fig .7



Why is it important to address?

Population growth is a vital problem facing Egypt in many fields. It is a source of several consequences:

- Urban congestion and overcrowding: As the infrastructure of cities and civil regions struggles to cope with the vast rapid growth that may be led to crush the traffic systems.
- Edge the resources: **Population growth** puts pressure on nature's resources, including the vital things to human life.

- Environmental degradation: **Population growth** has a role in spreading waste in the environment, deforestation, destroying natural habits, and climate change.

Egypt faces significant challenges due to its rapidly growing population, which has far-reaching implications for resources, infrastructure, and economic development. According to data from the *United Nations Population Fund (UNFPA)*, Egypt's population growth rate exacerbates pressure on essential services such as healthcare, education, and housing while also straining natural resources (UNFPA, 2023). As illustrated in **Fig. 7**, the population curve continues its upward trajectory, signaling the need for strategic planning and resource allocation to sustain future generations.

How can we address population growth?

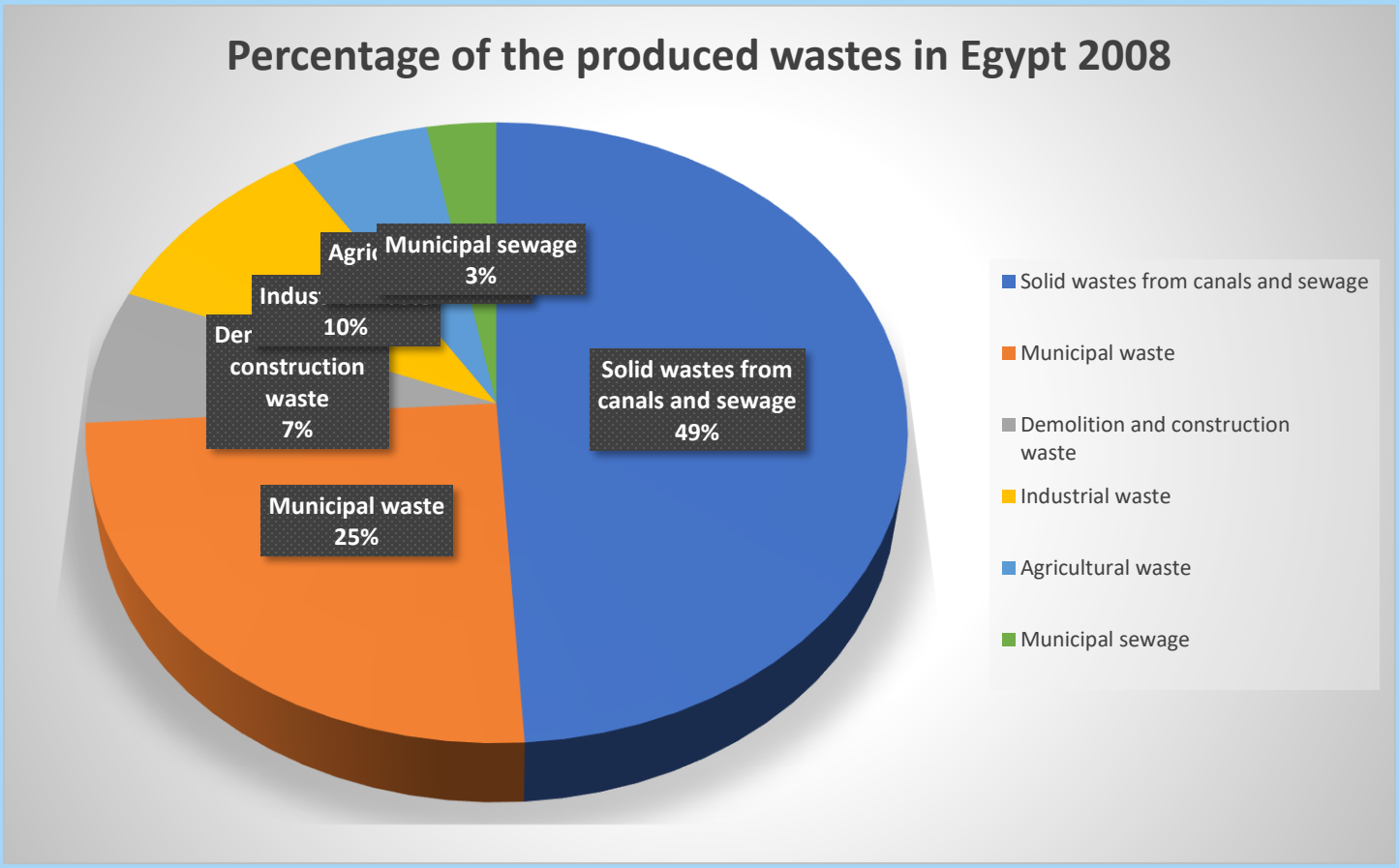
Improving connectivity and reducing congestion is a job that bridges take a place in. They can provide effective solutions to population growth problems by:

- ✓ Enhancing traffic flow: As bridges provide alternative paths, reducing travel time, decrement of traffic congestion on roads
- ✓ Connecting isolated areas: bridges are able to connect various regions.

V. Recycling garbage and wastes for economic and environmental purposes

Recycling garbage and wastes is a important for the environmental protection and economic efficiency. What is meant by recycle is reduce the

Fig .8



amount of garbage and wastes to landfills. This ensure a sustainable future for the country. According to a recent study by the Central Agency for Public Mobilization and Statistics, the percentages of the produced waste are as shown in **Fig (8)** we can observe that It is worth mentioning that, the type of waste of concern to the present study is the municipal waste. According to a study done by the National Environmental Action Plan of Egypt for the year 1992, household waste represented 68 per cent of the total municipal waste; waste from street sweeping was 12 per cent; waste from commercial processes was 11 per cent; waste produced from industrial establishments represented 5 per cent; and waste produced from institutions such as hospitals, hotels, and educational institutions was 4 per cent

Why is it important to address?

Using recycled resources will be less expensive, especially if we look for maintenance costs, which will preserve some budget for other stuff. Recycling garbage is crucial for several reasons:

1. Resource Conservation.

2. Energy Savings.
3. Reducing landfill waste.
4. Economic Benefits.
5. Environmental Protection.

How can recycle wastes in economic and environmental purposes?

Fig .9

Bridges can be manufactured using recycled garbage, as manufactured garbage can contribute to building a bridge, which is a particular way of reducing waste, solving urban congestion, helping to cut off population growth, and aiding in connecting arid areas.



Milik,S.M.(2021).Assessment of solid waste management in Egypt during the last decade in light of the partnership between the Egyptian government and the private sector [Thesis, the American University in Cairo]. AUC Knowledge Fountain. https://fount.aucegypt.edu/retro_etds/2473 How can we recycle wastes in economic and environmental purposes?

- ✓ Develop efficient recycling infrastructure.
- ✓ Raise public awareness to reduce contamination
- ✓ promote innovation in recycling technology.

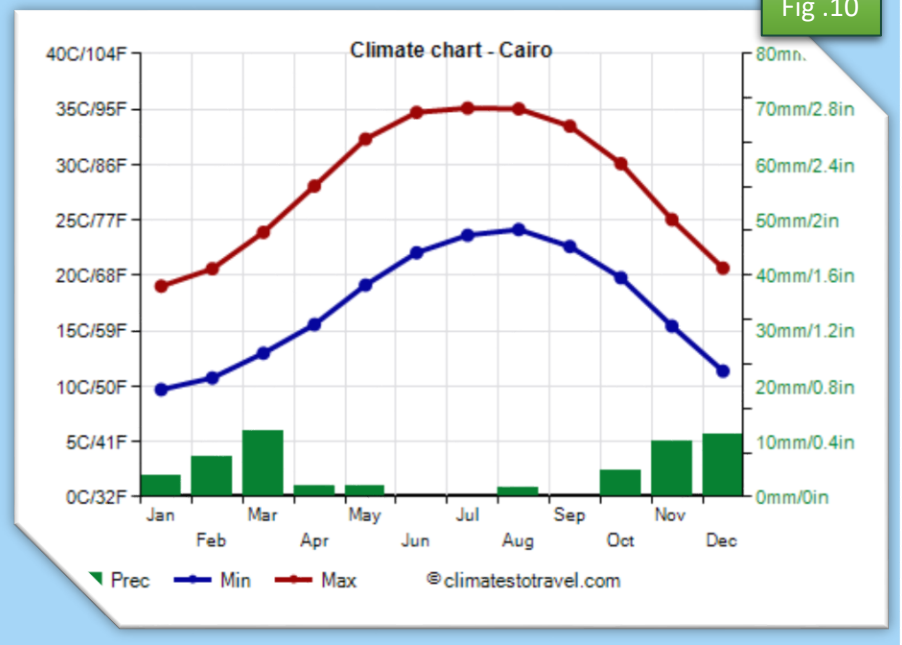
VI. Reducing the effect of climate change

Climate change refers to long-term shifts in temperatures and weather patterns. Such shifts can be natural; due to changes in the sun's activity, human activities have been the main driver of climate change, primarily due to the burning of fossil fuels like coal, oil, and gas, which affects not only Egypt but the entire world, and it is one of eleven grand challenges that Egypt faces. Climate change consequences:

- ∇ Rising sea levels: Egypt has a long coastline along the Mediterranean Sea, making it vulnerable to rising sea levels. This threatens low- lying areas particularly the Nile Delta, which is crucial for agriculture and habitation.

- ∇ Extreme weather events: Increased frequency and intensity of extreme weather events, such as heatwaves and heavy rainfall, can lead to flooding and disrupt livelihoods.
- ∇ Economic consequences: The economic impact of climate change can be profound, affecting food security, health, and overall stability. This can lead to increased migration and social tensions.

Fig .10



How can we reduce the effect of climate change?

1. Sustainable Materials: Using eco-friendly materials like recycled steel, concrete, or wood can lower the carbon footprint of building a bridge.
2. Reduce Traffic Congestion: Well-designed bridges improve traffic flow, reducing vehicle emissions and air pollution.
3. Encourage Green Transportation: Bridges that support bikes, pedestrians, and electric vehicles can promote cleaner, low-emission transport options.
4. Energy-Efficient Design: Some bridges use solar panels or other energy-saving technologies, reducing the need for electricity from non-renewable sources.

Egypt, Value, -- Prevalence of moderate or severe food insecurity in the total population (percent),” and “2020 / 2019-2021 2021 / 2020-2022 2022 / 2021-2023.Develop efficient recycling infrastructure.

https://carnegieendowment.org/research/2023/10/climate-change-in-egypt-opportunities-and-obstacles?lang=en

2. Problem to be solved

The world faces many problems, including urban congestion, and we can solve this problem or part of it by building a bridge, as the bridge addresses many of the problems facing the world, especially Egypt, the most important of which is urban congestion, because the bridge increases the capacity of roads, as the bridge provides additional lanes above and below

ground, which reduces pressure on the roads and accommodates more traffic. It also separates traffic lanes, as the bridge reduces the interference between different vehicles by creating multiple levels of traffic, as it separates pedestrians and cars heading in different directions. It connects areas more efficiently. The bridge also facilitates traffic between natural obstacles such as rivers or railways and helps stimulate internal trade.

If Problem Solved

If the problem is solved, it will have many positive effects in many economic, health and other areas.

1- Economic benefits

Increased productivity: When you reduce traffic congestion, this saves time for workers on the road, allowing workers to be more productive, which will boost the economy as companies will face much fewer delays in transporting goods.

2- Health benefits

Improved air quality because vehicles will travel shorter distances, which will reduce emissions, which will reduce respiratory and cardiovascular problems among populations suffering from poor air quality.

3- Environmental pollution

Reduction in Greenhouse Gas Emissions, Improved Air Quality, Encouragement of Sustainable Transport etc.….

If Problem Not Solved

If the problem isn't solved, it will have many negative effects in many economic, health and other areas.

1- Impact on travel

Urban congestion causes delays and increased travel times, especially during peak periods, which leads to frustration and increased stress among passengers.

2- Driver behavior

Urban congestion (traffic) leads to anger among drivers, which makes them display aggressive behavior, which leads to increased cases and lack of peace among people.

3- Environmental pollution

Environmental pollution will continue to worsen due to vehicle emissions (greenhouse gases)

4- Impacts on public transport users

It will continue to affect drivers and users, especially low-income individuals who are completely dependent on it.

3. Research

Topics for researching the problem

- Is the bridge a useful to connect areas and lands?
- What's the role of bridge in promoting regional economic integration?
- Why the bridge is the most effective solution for minimizing accidents, delays and risks?
- Can bridges provide a crucial pathway for emergency vehicles to reach the desired area quickly?
- How does the truss work to supply stability and steadiness for the bridge?
- Is population growth directly proportional to the arid areas' connectivity using ridged structures?
- Can bridge reduce the urban congestion on capitals and modern cities?
- Are the ferries being alternative way to cross water stream?
- Which places bridges should be constructed?

Topics for researching the solution

- How long should be the span of the bridge?
- What type of bridges are suitable? (Arch, Beam, suspension, Truss, Cantilever, etc.)
- Which type of trusses should be use?
- What perfect material used to construct inflexible bridge?
- How can we get palm tree midribs?
- What are the main parts of bridges?
- What are prior problems having the same conflicts?
- Can recycling garbage contributes in building bridge?
- How can we obtain energy from the motion of cars in the bridge?

4. Other Solutions Already Tried

Prior solutions that have been applied trying to solve this problem and its advantages and disadvantages: -

1. Van Buren Street bridge

Description

Fig .11



- Work began in June 1912 after a legal delay over funding was resolved, with Corvallis agreeing to pay a larger portion of the project.
- The Van Buren Street Bridge was a swing span.
- steel motor vehicle bridge spanning the Willamette River at Corvallis in the U.S. state of Oregon.
- Opened in 1913, the black-colored bridge was the first bridge across the river at Corvallis.

Bridge design

- A swing bridge, its main 249-foot (76 m)-span could swivel on its axis and open to allow river traffic to pass.
- Six people would rotate the key using a 17-foot (5.2 m) long wood rod that combined resemble a corkscrew.
- The operators could use two gears, one opened the span in 150 revolutions, while the other completed the job in 50 revolutions.
- When opened it provided 102 feet (31 m) of clearance on either side of the bridge.
- Van Buren Street Bridge was the last movable-span truss bridge constructed by the pin connection method located on the West Coast.
- Pins were used to connect the trusses, but the technique was abandoned after rivets came into usage.
- Overall, the bridge was 708 feet (216 m) long and was composed of three separate steel spans.
- The swing span was a Pratt through truss, and the next longest part a 171-foot (52 m) long Parker through truss span on the western end.

Advantages

1) **Efficient Traffic Flow**

- The Van Buren Street Bridge is a bascule bridge, meaning it can be raised to allow river traffic to pass.

2) **Strategic Location**

- The bridge is centrally located, connecting the Loop (downtown Chicago). This position provides easy access to major areas.

Disadvantages

1) **Traffic Delays During Lift Operations**

- When the bridge is raised to allow boat traffic, road traffic must come to a halt. This can lead to delays.

2) **Limited Capacity**

- The bridge was designed many years ago, so its capacity may not be suitable for today's increased traffic volumes

Citation

Rasmussen, B., Mejias, L., & Holder, K. (2010b). Martin Van Buren National Historic Site transportation analysis report. Portfolio 23110.

https://rosap.ntl.bts.gov/view/dot/9286/dot_9286_DS1.pdf

2. Tower Bridge

Description

Fig .12



- Tower Bridge is a Grade I listed combined bascule and suspension bridge in London, built between 1886 and 1894.
- It was designed by Horace Jones and engineered by John Wolfe Barry, with assistance from Henry Marc Brunel.
- The bridge crosses the River Thames near the Tower of London and is one of five London bridges owned and maintained by the City Bridge Foundation, a charitable trust established in 1282.

Bridge design

- feet the central opening span to be 200 feet clear with a height of 135 above Trinity high water when open, and a height of 29 feet when closed.
- The size of the piers to be 185 feet long and 70 feet wide.
- The length of each of the two side spans to be 270 feet.
- During construction a clear waterway of 160 feet wide had to be maintained for river traffic.
- The design of the bridge should be made to accord with the architecture of the Tower of London.
- The bridge was to be completed within 4 years from the passing of the Act.
- The bridge was to be opened at any time for the passage of any vehicles, regardless of any delays to land traffic.

Advantages

- Help the Capital's residents navigate the city faster amidst a rapidly growing population.
- Bridge consists of two towers, massive steel towers, that support bridge and make it stable.
- Ease road traffic while preserving access to a main stretch of the River Thames.
- located at a great vantage point.

Disadvantage

- Weight and Load Limitations: Despite its sturdy appearance, Tower Bridge has weight restrictions to protect its structure. Heavy vehicles like large trucks are not allowed.
- Traffic Congestion: Tower Bridge is a popular route for vehicles, pedestrians, and cyclists, leading to frequent congestion
- High Maintenance Costs: Due to its age and historical status, Tower Bridge requires regular and costly.

Citation

Tuit, J. E. (2011). The Tower Bridge: Its History and Construction from the Earliest Projects to the Present Time

https://doi.org/10.1680/ttbihacftdotepttpt.52611

3. Erasmusbrug

Fig .13

Description



- The Erasmusbrug (English: "Erasmus Bridge") is a combined cable-stayed and bascule bridge
- Construction began in 1986 and was completed in 1996.
- It crosses the Nieuw Maas in the center of Rotterdam, connecting the north and south parts of this city
- The bridge was named in 1992 after Desiderius Erasmus, a prominent Christian Renaissance humanist also known as Erasmus of Rotterdam.
- The Erasmus Bridge is Rotterdam's most important landmark and is even part of the city's official logo.

Bridge design

- Has a steel truss bridge, well-known for its strength and ability to handle heavy loads.
- Its truss design relies on a triangular framework that effectively distributes weight
- One of the bridge's standout features is its swing span, which allows it to pivot horizontally to let river traffic pass.
- The total length of the bridge is about 490 meters. This includes the spans that stretch across the Nile.
- The bridge is approximately 14 meters wide.
- The clearance height for river traffic, when the bridge is closed, is about 4.5 meters above the water level.
- The height of the bridge structure itself, including its trusses, can reach approximately 20 meters from the base to the top of the truss.

Advantages

- Iconic Landmark: The Erasmusbrug is a symbol of modern Rotterdam and a major tourist attraction, enhancing the city's global profile.
- Sustainability: The bridge facilitates non-motorized transportation like cycling and walking, promoting eco-friendly travel options
- Improved Connectivity: It connects the northern and southern parts of Rotterdam, improving transportation and accessibility for vehicles, pedestrians, cyclists, and trams.

Disadvantage

- High Cost: The construction cost was significant (approximately €165 million), and ongoing maintenance costs for such a complex structure are also high.
- Limited Lifespan of Movable Parts: The bascule mechanism requires frequent upkeep and may have a shorter lifespan compared to fixed sections of the bridge.
- Noise and Vibrations: Residents near the bridge have reported noise and vibrations, especially during heavy use or windy conditions.

Citation

www.architectuur.org. (n.d.). Erasmusbrug, Rotterdam, UN Studio | Architectuurgids. Architectuur.ORG.

https://www.architectuur.org/bouwwerk/478/Erasmusbrug.html

4. Mary MacKillop Bridge

Description

- The Mary MacKillop Bridge is a bascule bridge in Adelaide.
- In July 2005, Abi group was awarded a contract to build a railway bridge
- the adjoining Tom 'Diver' Derrick Bridge to carry the Port River Expressway over the Port River.
- The bridge was built as part of a project to divert the Dry Creek-Port Adelaide line away from the Rosewater loop and bypass the suburban network.
- It was opened on 1 August 2008 by Premier Mike Rann.

Fig .14



Design of bridge

- Primarily made of steel and concrete, with elegant railing designs and LED lighting to enhance night visibility and aesthetics.
- The bridge length measures approximately 150 meters in total, including approach ramps and the central crossing over the river.
- The main span of the bridge (the section that crosses directly over the river) is approximately 85 meters
- The bridge width ranges from 6 to 8 meters, allowing for pedestrian and cyclist traffic.

- The clearance height above the water (allowing for small boats and water activity) is approximately 4 to 5 meters depending on water levels.

Advantages

- Nighttime Usability: Integrated LED lighting makes the bridge accessible and visually appealing during night hours.
- Environmentally Friendly: Reduces the carbon footprint by encouraging eco-friendly transportation methods
- Safety: Offers a separate pathway for non-motorized traffic, ensuring safety away from vehicular roads.

Disadvantage

- Cost of Construction and Maintenance: High initial construction costs and ongoing maintenance expenses for lighting, cleaning, and structural upkeep.
- Weather Exposure: Being an open structure, it can become less usable during extreme weather conditions like heavy rain or strong winds.
- Potential for Overcrowding: During events at Adelaide Oval or in the city, the bridge might experience high pedestrian and cyclist traffic, leading to congestion.

Citation

MacKillops Bridge (Snowy River National Park, Snowy River Heritage River). (n.d.)

https://www.parks.vic.gov.au/places-to-see/sites/mackillops-bridge

5. Imbaba bridge

Fig .15

Description



- The Imbaba Bridge is a railway bridge located in Cairo.
- Egypt across the Nile River, about 935 kilometers (581 mi) downstream from the Aswan Dam.
- It is the only railway bridge across the Nile in Cairo.
- The current version was built between 1912 and 1924 by the Belgian firm Baume-Mar pent.
- The first iteration was constructed in 1891 to let trains cross the Nile west to the Giza Train Station.

Design of bridge

- Has a steel truss bridge, well-known for its strength and ability to handle heavy loads.
- Its truss design relies on a triangular framework that effectively distributes weight
- One of the bridge's standout features is its swing span, which allows it to pivot horizontally to let river traffic pass.
- The total length of the bridge is about 490 meters. This includes the spans that stretch across the Nile.
- The bridge is approximately 14 meters wide.
- The clearance height for river traffic, when the bridge is closed, is about 4.5 meters above the water level.
- The height of the bridge structure itself, including its trusses, can reach approximately 20 meters from the base to the top of the truss.

Advantages

- Dual-Purpose Design: Its ability to accommodate both rail and road traffic makes it a multipurpose infrastructure.
- Durable Construction: Built primarily of steel, the bridge has stood the test of time, withstanding heavy use and environmental exposure for nearly a century.

Disadvantage

- Limited Capacity: The bridge's width, designed for its time, may struggle to handle modern traffic volumes, leading to congestion, especially during peak hours.
- Environmental Impact: The bridge's maintenance and use, especially with heavy trains and vehicles, contributes to pollution and carbon emissions in the surrounding area.

Citation

Al-Azhar University Civil Engineering Research Magazine (CERM) Vol. (41) No. (3) July, 2019

https://www.azharcermjournal.com/CERMF1907/P19-07-22.pdf

Generating and Defending a Solution

1. Solution and Design Requirements

Solution requirements

Any solution must have requirements to be successful.

I. Availability

Must the solution be available in Egypt and not imported from another country with a lot of money. The solution should depend on Egypt resources. The solution should be available for everyone. The solution should be available in order to be used easily by everyone in order to succeed in solving the problem because if a solution is effective and good but not available it won't be used, leading to the failure of the solution.

II. Eco – friendliness

For a solution to be eco-friendly, many requirements should exist in the project. Must use alternatives energy resources to avoid the negative impacts of fossil fuels like coal and petroleum. As Egypt is a country that is extremely affected by pollution, finding an eco-friendly solution would be valuable. The materials that we will use must be eco-friendly and do not affect the environment and living organisms.

III. Durability

durability should be studied critically; the durability of the project means sustainability and enduring of the project along with long-lasting periods of time. The project should have the ability to withstand any conditions, weather, or excessive use. Because the project would be used on a large scale, durability should be guaranteed.

IV. Sustainability

The sustainability of the project is a very essential factor to determine whether the project could be applied in real life or not. The project should sustain the surrounding conditions of the present and meet the needs of future generations. Accordingly, it would still give the same advantage for the country throughout long years in the future.

Design requirements

Any prototype must have requirements to be successful.

You must follow ALL School Safety rules.

- The scale of the constructed prototype is 1: 50
- The total length of the bridge ranges between 40 and 50 meters.
- The bridge must have a total of two lanes of vehicles plus two lanes for pedestrians.
- The width of the traffic lane is 3.5 m while the width of the pedestrian lane is 1.5 m.
- An island of 1 m should be provided in the middle width of the bridge.
- The clearance between the water level and the lowest point of the deck of the bridge is 5 m.
- The bridge should allow the passage of ships that require a minimum width of 12m to a maximum width of 14m
- a minimum of 10m height over water surface (table or ground for your prototype) through the movable part of the bridge.
- The mechanism of opening and closing the movable part of the bridge could be either mechanical or hydraulic.
- (You can use your hands to operate the mechanism, but you cannot use your hands or a hand tool to directly open or close the bridge.)
- Our bridge must carry a simulated "traffic load" of a minimum 10 kg mass (not scaled) placed on the structure at the mid-point of the largest fixed span. The maximum deflection, at the same point, does not exceed 5 cm (not scaled) without collapse.
- Measure and present a graph of the relationship between the load (increasing from zero to the maximum) and the corresponding deflection at the same point.
- The maximum weight of the whole prototype should not exceed 20 Kg.
- I choose these requirements because in my opinion these requirements will be suitable for the solution.

2. Selection of Solution

First, Egypt has many problems that must be solved, and these problems are called Egypt's Grand Challenges. We have reached the solution to these problems.

The solution idea:

After thinking a lot, we reached the solution; the solution is a bridge. A bridge can help us to reach the solution to many problems such as arid areas, climate change, urban congestion, and recycling waste.

How the bridge can help

1. We can build the bridge from recycled waste, which can help us to reduce waste.
2. We can build a bridge to connect two areas, one full of people and the other empty; that will help address the challenge of arid areas.
3. When building the bridge, it helps alleviate congestion on nearby roads, improving overall traffic management in the area.
4. Addressing these problems with the bridge will help reduce climate change and global warming.
5. It can connect populated and arid areas, promoting sustainable land use.
6. The solution aids in combating climate change and global warming.
7. Building the bridge can create job opportunities and boost the local economy.
8. It contributes to disaster resilience by providing an evacuation route in emergencies.

3. Selecting of Prototype

The proposed bridge design we recommend is a single-leaf bascule bridge.

This type of bridge has only one moving section, which is relatively like a seesaw; two stationary sections and one movable are included in the suggested bridge design. Two pillars are connected to the bridge, and a guide pulley is mounted inside the pillars to act as a track for the cable.

The mechanical system of the bridge relies on electric motors; these motors are accurately controlled by an Arduino microcontroller board, which operates them through an H-bridge.

The bridge also includes mechanisms that provide mechanical advantage. This mechanical advantage is mainly constructed by pulleys, drums, and cables. This mechanical advantage can reduce and change the force needed to lift the deck and increase the torque.

This solution meets the specified requirements by ensuring structural integrity and offering the desired outcome. Our bridge can improve traffic flow and cut down travel duration, establishing a direct route for commuters, guaranteeing the safe transmission of ships, helping in addressing arid areas by improving access to water resources in these areas, and promoting urban expansion. We also provide a 3D model the simulates the bridge shown in fig. 16

Fig .16



Constructing and Testing a Prototype

1. Materials and methods

Table .1

Items	Palm tree midribs	Wood glue	Screws	Hinges	Arduino board	Motors
Quantity	120 Midribs	20 – 30 Screws	1 Flask	3 Hinges	1 Arduino board	2 Motors
Description	They are strong, stiff parts in palm trees. They hold the leaf and can be used for making things.	Small metal rods. They are used to hold things together by twisting them into wood, metal, or other materials.	Sticky liquid used to join pieces of wood together.	Metal parts that let doors, lids, or gates swing open and close.	An Arduino board is a small, programmable device used to control electronics. as it can read inputs.	Machines that turn electricity into movement. They make things spin or move, like fans or wheels.
Usage	It will be utilized to fabricate various components of the bridge.	Attach the superstructure and the hinges with the substructure.	Bond the stalks together and provide strength and durability.	Allow the movable part to rotate around fixed axis.	A board used to control electric components (dc motors, limit switches, IR receiver)	Lifting the movable part and towing the polyester rope.
Cost	Free (wasted)	FABLAB	40 LE	Free	500 LE	315 LE
Source	Local palm trees	Local store	Local store	Local store	Online store (Makers)	Online store (Makers)
Picture						

Table .2 Items	IR (Receiver & Remote)	Jumper wires	H-Bridge	Battery	Nylon fishing rope	Limit switches
Quantity	1 IR	25 – 30 Wire	1 L293D	1 Battery	2 Meters	2 Limits
Description	An IR receiver detects infrared signals from a remote control and converts them into commands for devices.	They are short wires used to make connections between different parts of a circuit, often in electronics projects.	An H-bridge is a circuit used to control the direction of a motor. It allows the motor to rotate forward, backward, or stop by switching the flow of current.	A battery stores energy and supplies power to devices.	Nylon rope are strong, flexible fibers used for making ropes, they're durable and resistant to wear	Limit switches detect an object's position and stop or control movement.
Usage	It will be utilized to fabricate various components of the bridge.	Attach the superstructure and the hinges with the substructure.	Bond the stalks together and provide strength and durability.	Allow the movable part to rotate around fixed axis.	A board used to control electric components (dc motors, limit switches, IR receiver)	Lifting the movable part and towing the polyester rope.
Cost	Free (wasted)	FABLAB	40 LE	Free	500 LE	315 LE
Source	Local palm trees	Local store	Local store	Local store	Online store (Makers)	Online store (Makers)
Picture						

Table .3 Items	Lab coat	Latex gloves	Goggles
Usage	To protect the body of the person working.	To protect against the dirt and microbes that the person working on the project might get exposed to	To protect the eyes of the person
Picture			

The construction of the prototype requires sticking to the design requirements and configuring the electric circuit as illustrated in **fig. (5)**. Primarily, the method of how the movable part is going to allow the safe transmission has been chosen, followed by specifying the dimensions through modeling a 3D design that simulates the prototype as shown in **fig. (3)**.

Steps of installation:

I. Cutting down palm trees into equally divided limbs while utilizing a machete, the bridge was mainly sliced into three parts: two movable parts and one fixed. The three parts are the same in width ($26 \pm 0.5\ cm\ in\ width$) but the first part, unlike the other two parts ($35 \pm 0.5\ cm\ in\ length$), the other two parts are ($30 \pm 0.5\ cm\ in\ length$).

II. The palm tree midribs were inserted inside each other, making a hole inside one and fitting the other inside, as shown in **fig. (4)**. This process happened with the aid of the FABLAB using a driller.

III. We started adhering the parts to each other via screws and wood glue, forming the backbone of our bridge, like: (deck - abutments - piers - pillars); then building the truss comes after; we decide to construct a Pratt truss; we talk into

account building a base that serves as a platform for the motors and compose a stage of $(2 \pm 0.3\ cm\ in\ width)$ so when the bridge lowers, it can lie upon a base.

IV. Motors, ropes, drums, guide pulleys, and hinges have been installed precisely.

V. Arduino circuit and its components, including the H-Bridge motor driver, have been properly assembled and configured; this wouldn't be easy without setting up a circuit model.

2. Test plan

The bridge mainly has design requirements of having a great span of $\mathbf{95 \pm 1\ cm}$ in length, and the $\mathbf{28\ cm}$ span of the bridge provides clearance for the marine traffic to navigate through. It also has a weight of $\mathbf{300\ g}$. It has a capacity to carry a weight of up to $\mathbf{10\ kg}$. This draws attention to its durability and unwavering stability. Additionally, it features two lanes specified for the vehicle's width of $\mathbf{7 \pm 0.5\ cm}$ for one and the other two lanes for the pedestrian's width of $\mathbf{3 \pm 0.3\ cm}$, with a separating island in the center with a width of $\mathbf{2 \pm 0.3\ cm}$.

This solution meets the specified requirements by ensuring structural integrity and offering the desired outcome. Our bridge can improve traffic flow and cut down travel duration, establishing a direct route for commuters, guaranteeing the safe transmission of ships, helping in addressing arid areas by improving access to water resources in these areas, and promoting urban expansion.

Test plan to estimate deflection:

1. Every process in the test plan has been done through safety regulations and right precautions as putting on masks and gloves, waring lab coats, avoid getting hurt while using any dangerous equipment.
2. We drop by degrees a weight, over time we increase the weight; therefore, the distance between the bridge after putting weight and before increase, then measuring this difference utilizing a vernier caliper. This operation was supervised by the mechanics teacher in the mechanics lab.
3. Documenting data we observed in the form of tables.

3. Data collection

A detailed representation of the weight-deflection relationship is provided in the graph **fig .18** and table. **table. 4** this means that the bridge can lift up to 10 kg. with small deflection of

0.8 mm this is a bright result our bridge provided, our bridge also provides a suitable opening and closing speed for the vehicles and pedestrians this is represented in graph **fig. 17**

Fig .17

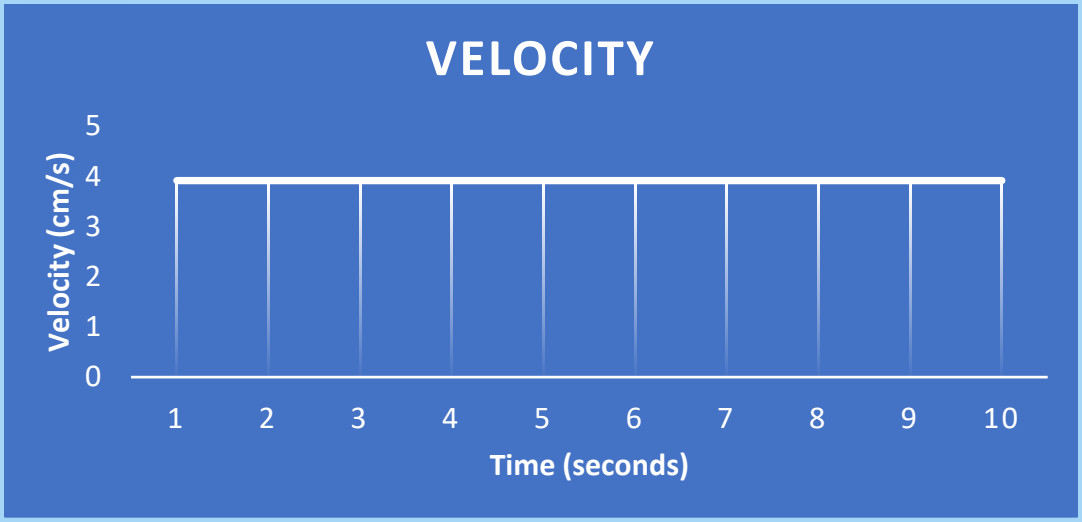


Table .4

Weight (Kg)	Deflection (mm)
0	0
0.5	0.1
1	0.1
1.5	0.2
2	0.3
2.5	0.3
3	0.4
3.5	0.4
4	0.4
4.5	0.5
5	0.5
5.5	0.5
6	0.5
6.5	0.6
7	0.6
7.5	0.6
8	0.7
8.5	0.7
9	0.7
9.5	0.8
10	0.8

Fig .18

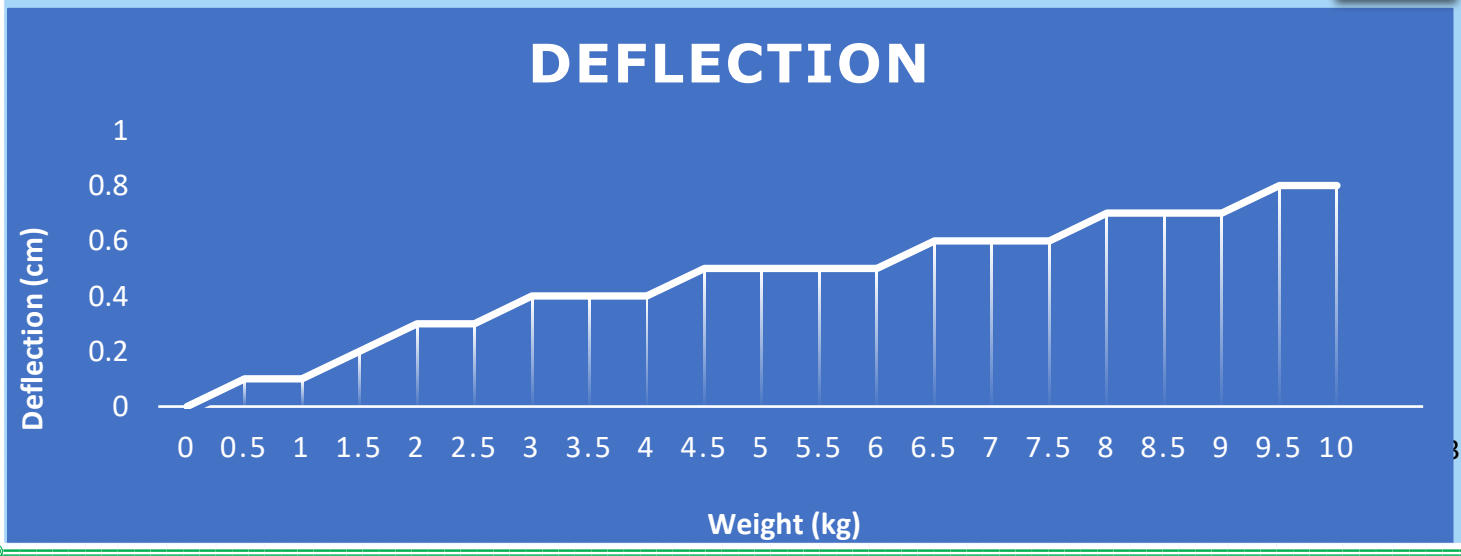
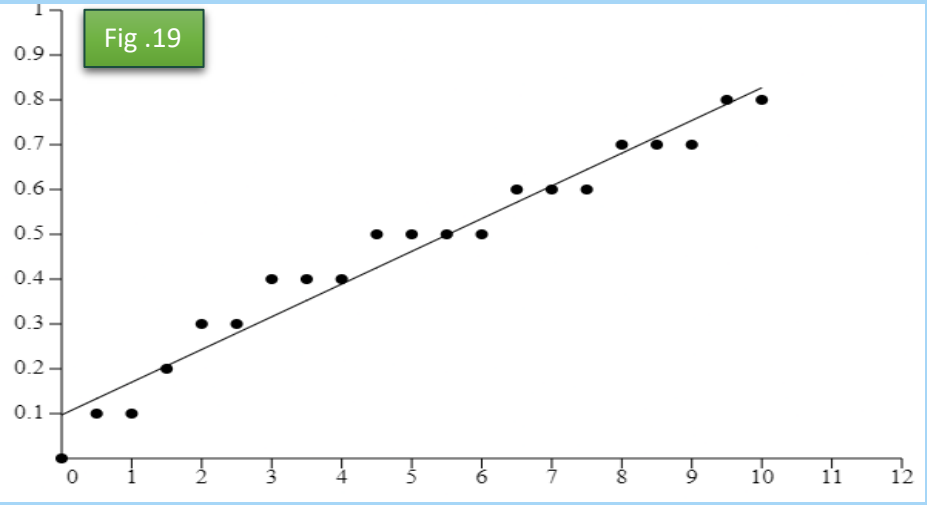


Fig .19



this is a graph **fig. 19** that provide the linear regression of the wight the prototype can bear.

Also, a table represents the pressure that the bridge exposed is been shown in **table .5**

Table .5

Weight (Kg)	**Force** (N)	**Area** (m^2)	**Pressure** (Pa)
1	9.8	0.066	148.48
3	29.4	0.066	445.45
5	49	0.066	742.42
7	68.6	0.066	1039.39
9	88.2	0.066	1336.36

Evaluation, Reflection, Recommendations

1. Analysis and Discussion

Over the period of the three years in STEM Schools, Egypt faces eleven major challenges that require STEM solutions. This semester's project aims to solve five major issues facing Egypt (improving the use of arid areas, deal with urban congestion, recycled garbage, and wastes for economic and environmental purposes, and reduce the effect of climate change) These problems will increase in the future for the world, especially Egypt. Therefore, a solution is required to be found to prevent us from having more effects of these problems in the future.

To solve these problems and produce a scientific solution, we used the (EDP) process.

- ❖ Start with, the problem was identified by discussing related topics such as (causes of urban congestion, increasing arid areas in Egypt and Prior solutions that aim to solve like these problems).
- ❖ Second, in terms of solutions, multiple topics were examined to determine the best way to solve these problems.
- ❖ After careful study, constructing a bridge was the best way to solve the matter.

The main **design requirements** must be in the **perfect** solution:

a. The **total length** of the bridge is **90 cm**.
b. The bridge must have a total of **two lanes** of traffic vehicles plus **two lanes** for pedestrians. The width of the traffic lane is **7 cm** while the width of the pedestrian lane is **3 cm**. **An island** of **2 cm** should be provided in the middle width of the bridge.
c. The **clearance** between the water level and the lowest point of the deck of the bridge is **10 cm**.
d. The bridge allows **passage of ships** that require a maximum width of **28 cm** and a minimum of **20 cm** height over the water surface.
e. The bridge carries a simulated "**traffic load**" of a minimum **10 kg** mass placed on the structure at the mid-point of the largest fixed and

maximum of **67 kg** mass. The maximum deflection, does not exceed 5 cm

f. The maximum weight of the whole prototype should not exceed **20 Kg**.

Our prototype has some **advantages** like:

- Its ability to perform its function of transferring vehicles and allow the traffic flow of ships.
- It is constructed out of inexpensive, strong, and common (not rare) material as palm tree midribs.
- It uses an Arduino system to control the movement of opening and closing the bridge.

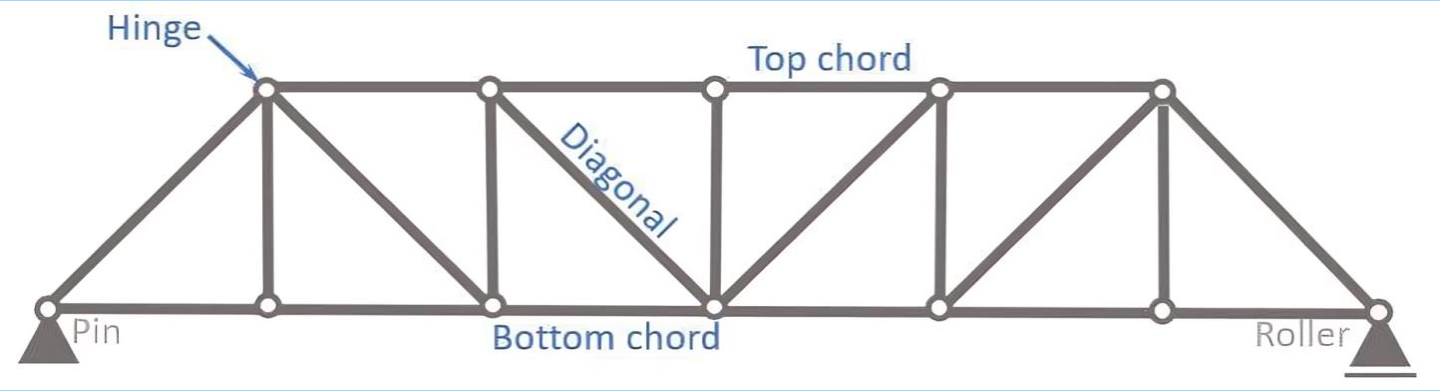
It also has some **restrictions** as:

- Its size is very big and troublesome carrying it.
- Due to its distinctive design, it requires high skill in its construction, which is difficult to find it.
- There is a little difficult to design the movable part and control it opening and closing due to its stage that acts as carrier for the movable part.

Truss structure:

Everyone knows that the best solutions, especially for experts, are obtained after many tests and trials. After researching the best truss of the bridge that we would employ, we found that **Pratt truss** is the most suitable truss for the bridge. Shown in **fig. 20**

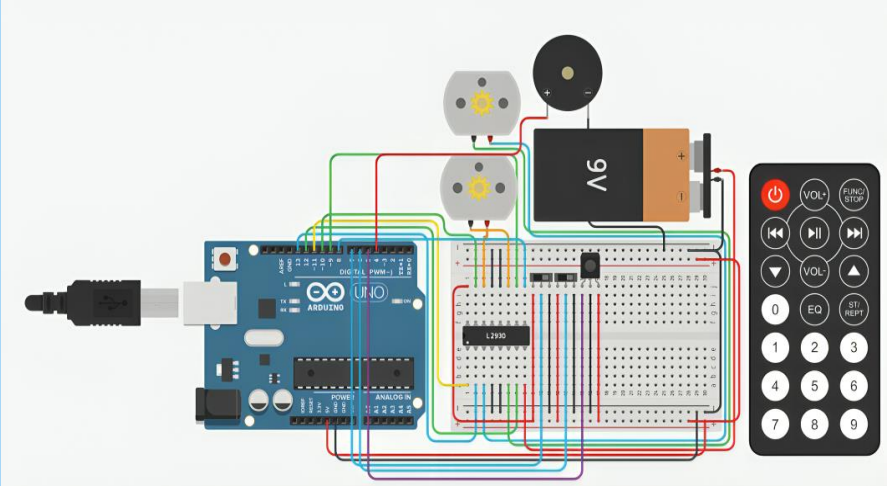
Fig .20



The Arduino circuit:

Fig .21

The prototype is equipped with an Arduino system that manipulates the DC gear motors through an H-Bridge, enabling the motors to rotate in both directions and adjust speed. Limit switches are included to indicate the boundary of motor movement. Additionally, the system includes a buzzer, LEDs, a battery, an IR remote, and an IR receiver, ensuring optimal control with high efficiency. The circuit is in fig. (21)



The **force** is important in bridges because it enables the movement of the bridge's movable part, allowing the bridge to handle heavy loads. A series of steps should be followed to calculate forces of motors. Firstly, by calculating the **torque**, the torque generated by the motors is crucial for overcoming the gravitational force acting on the movable part. The motors torque is used to turn the bridge and lift the movable part. The more torque the motors provide, the faster and easier the bridge can be lifted. After many calculations we obtained that the force acting on the movable part to open equal $\mathbf{5.765\ N}$

For a bridge, **angular velocity** is vital to determining the speed of rotation. Around the axis how quickly the bridge opens and closes. The angular velocity of the bridge is $\mathbf{0.1305\ rad/sec}$.

2. Recommendations

There is no perfect prototype, and our prototype is no exception. We recommend some features and characteristics that can have a significant impact on the bridge and that can help improve the application and efficiency of our project. For those who want to take some steps in improving this project, they can do the following:

1. Try to use renewable energy, especially solar panels, to power the motors as this will be better than using batteries. This will save as much electricity as possible.

2. Try to use stronger materials than palm ribs, we suggest using cement and steel. Since palm ribs are not a strong enough material, consider using recycled materials.

3. We suggest using ultrasonic sensors instead of using a remote control. This can create an automatic operating system that can only be opened when ships pass.

4. Install a drainage system to direct rainwater away to prevent water from damaging parts of the bridge.

5. Use smart cameras to monitor traffic or road conditions; this can be effective in recording what is happening on the road and shipping water paths; this can also provide information about incoming ships and water traffic.

6. Provide green spaces. This can be done by planting green areas in pedestrian walkways; this can reduce the carbon dioxide emissions from cars and vehicles.

7- Design an improved prototype:
Considering that it has performed very well and reached the targets, it would be wise to focus on improving different areas such as performance efficiency, cost and strength so that the prototype is practically applicable in the real world.
On the other hand, a quick and easy way to improve the reliability of the information obtained is to create more accurate measurement tools.
8- Future endeavors: Imagine new features that can be included in the prototype such as artificial intelligence or the Internet of Things as these technologies drive technology with the aim of improving it.

It is certainly necessary to look for the use of the prototype in other sectors or markets to increase the impact of the project.

9- Tips for future project teams:
Document each stage of the design and testing carefully with the clear goal of providing a stimulating impact on the next development of the product.

10- List best practices and the most important problems you faced and what measures you used to address them.

11- Building skills: We also work to achieve this plan of action as a possibility to develop scientific and engineering skills along with focusing on them and thus the progress of the program will be achieved.

3. Learning Outcomes

PH 1.01	**Measurement errors occur when we measure the length and width of the bridge and during the test plan when we measure the deflection.**
PH 1.02	We used the free body diagram to analyze the forces acting on the bridge and also during the test plan and at the joints to analyze the forces acting on them to bring the truss into static equilibrium.
PH 1.03	There was a change in motion as the bridge opened and closed from rest to motion and vice versa.
PH 1.04	We used the weight formula [w=mg] to determine what force would act on the body during the test plan and make the bridge bear more than it does.
PH 1.05	Torque law [f*r] was used to calculate the appropriate torque for the motor to lift the bridge. We also used a simple machine (pulley) to place the rope and the angular velocity and angular acceleration occurred as the bridge opened and closed.

MA 1.02	Statistics was used to organize data such as using a histogram of a population growth problem and linear regression in deflection.
MA 1.04	In reality, there will be a link (magnification) between the prototype and the bridge.
CH 1.01	Scientific methods helped us to identify the problem and find the best way to build a bridge that addresses most of the problems it can address in the best possible way.
ES 1.03	building materials was used to build the bridge in reality such as cement, steel, etc.
BI 1.01	The bridge will help reduce diseases (infectious and non-infectious) by reducing traffic congestion on the road and thus reducing the transmission of infectious diseases. The roads will also be less crowded and thus reduce stress and nervousness, which reduces non-infectious diseases such as pressure and others. It will also make the road faster to reach hospitals.
ME 1.01	Graphs was used such as a velocity-time graph to describe the speed of the bridge opening and closing, but the velocity was converted to angular velocity by dividing by r.
CS	Sketchup was used to create a design for the bridge. Flowchart was used to organize ideas and describe how information will be accessed.
E	We used English language skills (how to write an essay) when we wrote a poster or portfolio.
PH 1.06,7	Hooke's law helped us choose the type of string, that the string doesn't break when it raises the movable part, and the stress (f/area) would be used when the parts of the bridge separate.
MA 1.03	When the stress was calculated, cross section would be used, because stress = f/area, the force would be internal.

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Message of thanks and appreciation

In the end, we are grateful of Allah for helping us on our journey. We would like to thank all the teachers at QENA STEM School, especially professors Mohamed Moktar (Capstone teacher), Mahmoud Abdallah (G-10 Leader Capstone) and Ahmed Seif (G-10 General Capstone). We are grateful to our friends in G-11 and G-12 for their help and for providing us with the information we need. We appreciate Engineer/Ahmed Elnaggar and Engineer/Mohamed who supplied us with more information during our Journey in the Capstone project.